Long-range spectral statistics of the Rosenzweig-Porter model

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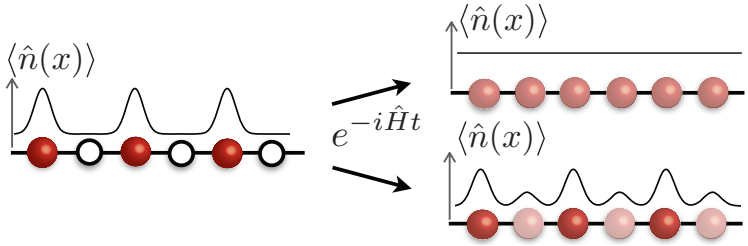
Outline

- **Introduction**
  Non-ergodic phases in complex quantum systems
  Rosenzweig-Porter model
- **Unitary Rosenzweig-Porter model**
  Dyson Brownian motion
  Sampling algorithm
- **Long-range spectral statistics**
  Spectral form factor
  Probing the ergodic-fractal transition
- **Conclusions and outlook**
- **Appendix: some results on other topics**

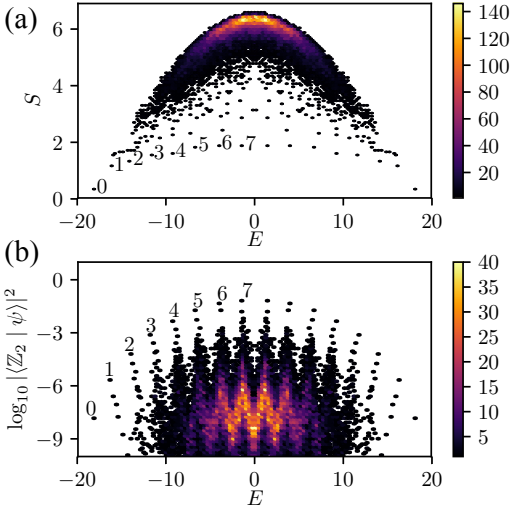
Non-ergodic phases in complex quantum systems

- **Quantum many-body systems typically thermalize**
  Eigenstate thermalization hypothesis
  Non-integrability

- **Exceptions do exist**
  Quantum many-body scarring
  Many-body localization
  Integrability



Abanin, Altman, Bloch, Serbyn, RMP 2018



Turner, Michailidis, Abanin, Serbyn, Papić, Nat. Phys. 2018

Non-ergodic phases in complex quantum systems

- **Fractality of eigenstates**

$$\text{IPR} = \sum_n |\langle n|\psi\rangle|^4 \sim N^{-d}$$

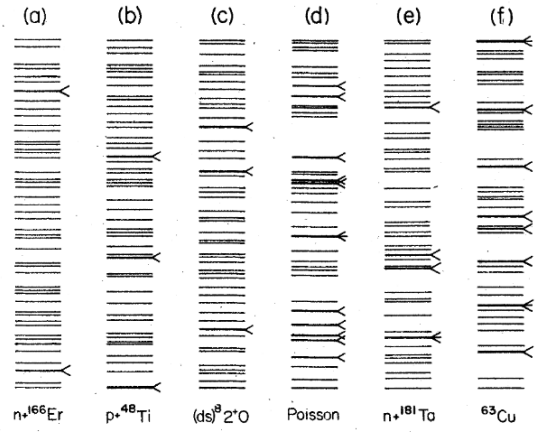
$d = 1$: ergodic
$d = 0$: localized
$0 < d < 1$: fractal

- **Level statistics**

$$r_i = \min\left(\frac{E_{i+2} - E_{i+1}}{E_{i+1} - E_i}, \frac{E_{i+1} - E_i}{E_{i+2} - E_{i+1}}\right)$$

$\langle r\rangle \approx 0.600$ (Wigner-Dyson, ergodic)
$\langle r\rangle \approx 0.386$ (Poissonian, localized)

- **Many other probes exist**



Brody, Flores, French, Mello, Pandey, Wong
RMP 1981

Rosenzweig-Porter model

- **Rosenzweig-Porter model**
  Random matrix model (single parameter)

$$H = \begin{pmatrix} X_1 & & \\ & \ddots & \\ & & X_N \end{pmatrix} + \frac{1}{\sqrt{N^\gamma}} \begin{pmatrix} \text{GUE} \end{pmatrix}$$

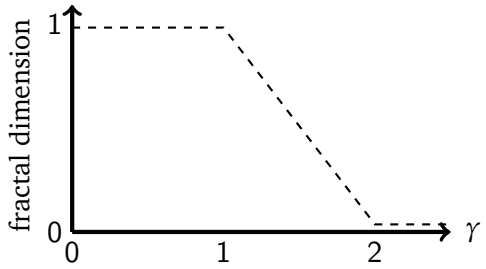
$\gamma \geq 0$: parameter (next slide)

$X_n \sim \mathrm{N}(0,1)$ Gaussian distributed

- **Gaussian unitary ensemble (GUE)**
  $P(M) \sim e^{-\frac{1}{2}\operatorname{Tr}(M^2)}$
  No preferred basis: $P(U^\dagger M U) = P(M)$
  Independent elements (second-next slide)
  Complex-valued elements (broken time-reversal symmetry)

Rosenzweig-Porter model

- **Competition between "ergodic" and "localized" terms**



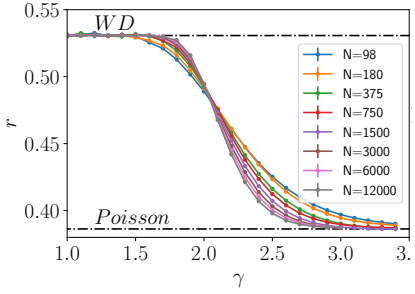
$0 < \gamma \leq 1$: ergodic

$1 < \gamma \leq 2$: fractal

$\gamma > 2$: localized

$\boxed{\text{Two transitions}}$

- **Transition between Poissonian and Wigner-Dyson level statistics**



Pino, Tabanera, Serna, J. Phys A: Math. Theor. 2019

Fractal – localized transition "invisible"

Recent modifications and generalizations

- **Survival probability in Generalized Rosenzweig-Porter random matrix ensemble**
  De Tomasi, Amini, Bera, Khaymovich, Kravtsov, SciPost Phys. 2019
- **Fragile extended phases in the log-normal Rosenzweig-Porter model**
  Khaymovich, Kravtsov, Altshuler, Ioffe, PRR 2020
- **Lévy-Rosenzweig-Porter random matrix ensemble**
  Biroli, Tarzia, SciPost Phys. 2021
- **Emergent fractal phase in energy stratified random models**
  Kutlin, Khaymovich, SciPost Phys. 2021
- **Circular Rosenzweig-Porter random matrix ensemble**
  Buijsman, Bar Lev, SciPost Phys. 2022
- **Non-Hermitian Rosenzweig-Porter random-matrix ensemble**
  De Tomasi, Khaymovich, PRB 2022
- **Replica approach to the generalized Rosenzweig-Porter model**
  Venturelli, Cugliandolo, chehr, Tarzia, SciPost Phys. 2023
- **Spectral statistics of a minimal quantum glass model**
  Barney, Winer, Baldwin, Swingle, Galitski, arXiv:2302.0070 (2023)
- **Tuning the phase diagram of a Rosenzweig-Porter model with fractal disorder**
  Sarkar, Ghosh, Khaymovich, arXiv:2305.13370 (2023)
- **Ergodicity-breaking phase diagram and fractal dimension in models with correlated disorder**
  Roy, Basu, Khaymovich, arXiv:2307.03085 (2023)

Unitary Rosenzweig-Porter model

- **Rosenzweig-Porter model (reminder)**

$$H = \begin{pmatrix} X_1 & & \\ & \ddots & \\ & & X_N \end{pmatrix} + \frac{1}{\sqrt{N^\gamma}} \begin{pmatrix} & \text{GUE} & \end{pmatrix}$$

- **Gaussian unitary ensemble**

$$M = \frac{1}{2}(W + W^\dagger)$$

$$W_{nm} = u_{nm} + iv_{nm}, \quad u_{nm} \sim \mathsf{N}(0,1), \quad v_{nm} \sim \mathsf{N}(0,1)$$

- **Dyson Brownian motion interpretation**

$$H(0) = \mathrm{diag}(X_1, X_2, \ldots, X_N)$$

$$H(t + dt) = H(t) + \sqrt{dt}M$$

Rosenzweig-Porter model results at $t = N^{-\gamma}$

Circular Dyson Brownian motion

- **Dyson Brownian motion**

  $H(0) = \mathrm{diag}(X_1, X_2, \ldots, X_N)$

  $H(t+dt) = H(t) + \sqrt{dt} M$

- **Circular Dyson Brownian motion**

  $U(0) = \mathrm{diag}(e^{i\theta_1}, e^{i\theta_2}, \ldots, e^{i\theta_N})$

  $U(t+dt) = U(t)\, e^{i\sqrt{dt} M}$

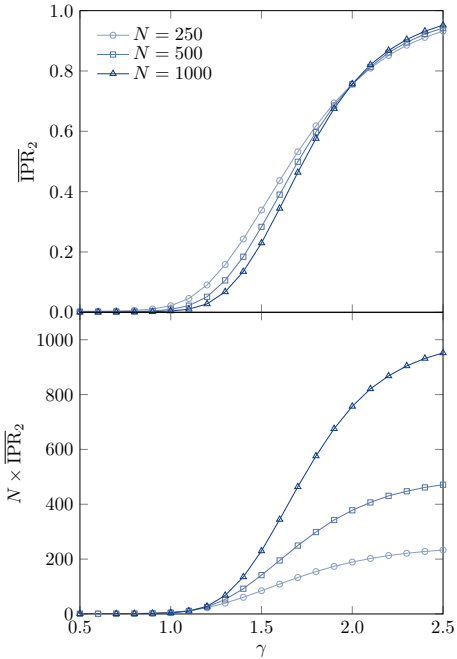
  Dyson, J. Math. Phys. 1962

- **Unitary Rosenzweig-Porter model**

  $\theta_n \sim \mathrm{UNIF}(-\pi, \pi]$

  Evolution up to $t = N^{-\gamma}$

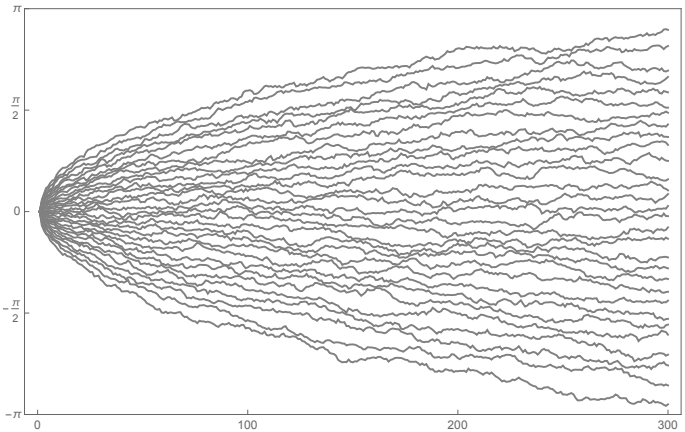
  Buijsman, Bar Lev, SciPost Phys. 2022



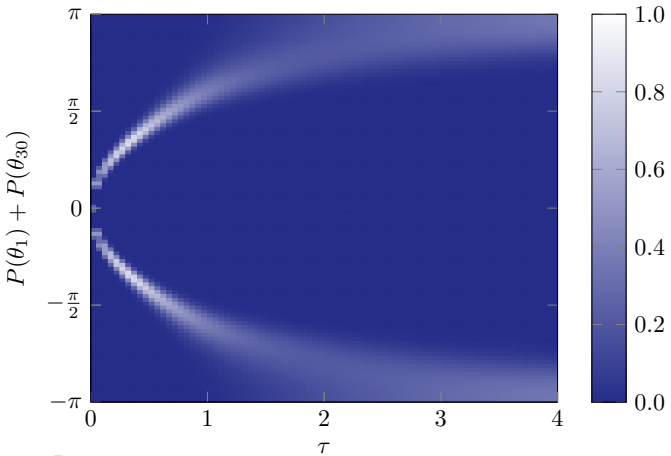
Problem: numerical evaluation is computationally very expensive

An improved algorithm

- **Evolving over a single time-step**
  Generalization of algorithm to generate random unitary matrices (Mezzadri, 2006)
  Opens possibility to study large-dimensional matrices



Forrester, Kieburg, Li, Zhang, arXiv:2206.14950 (2022)



Buijsman, in preparation

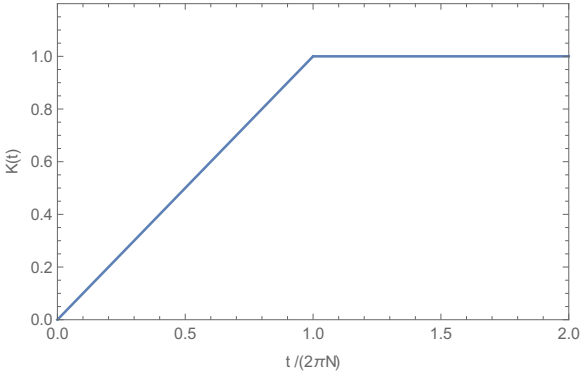
Long-range spectral statistics

- **Can the ergodic – fractal transition be probed by spectral statistics?**
  Spectral form factor (long-range) correlations
  Unitary model: no unfolding required

- **Spectral form factor**
  Fourier transform of two-point correlation function
  Thouless time: lowest time from which sff matches the random matrix evaluation
  Šuntajs, Bonča, Prosen, Vidmar, PRE 2020

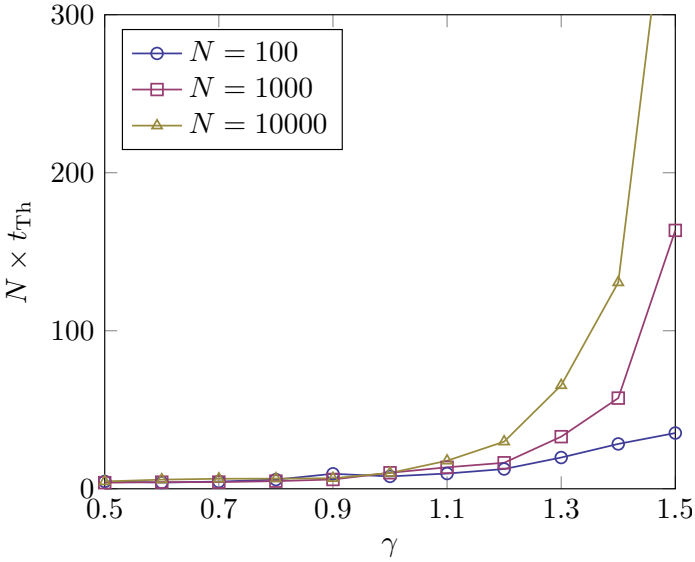
- **Circular unitary ensemble**

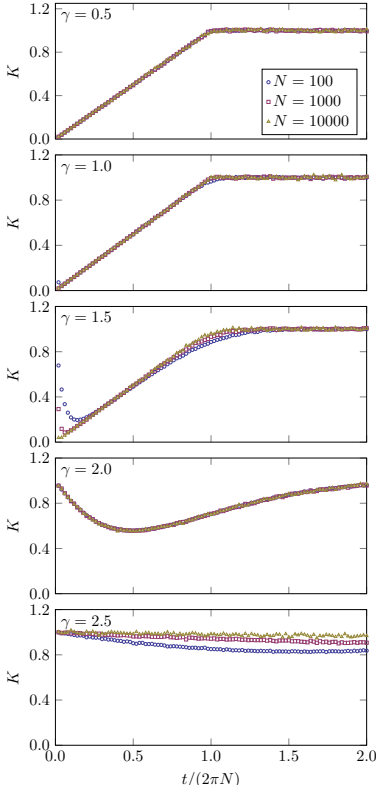
$$K(t) = \begin{cases} \frac{t}{2\pi N} & \text{if } |t|/N \leq 2\pi, \\ 1 & \text{if } |t|/N > 2\pi. \end{cases}$$



Long-range spectral statistics

- **Numerical evaluation**
  Probing of ergodic – fractal transition ($\gamma = 1$)
  "Critical" statistics at $\gamma = 2$





Conclusions and outlook

- **Take-home messages**
  Unitary Rosenzweig-Porter model
  Probing of ergodic – fractal transition through spectral statistics

- **Possible next steps**
  Further characterization of spectral statistics
  "Critical" statistics at $\gamma = 2$

- **Acknowledgements**

Appendix: some results on other topics

- **Quantum many-body scars**

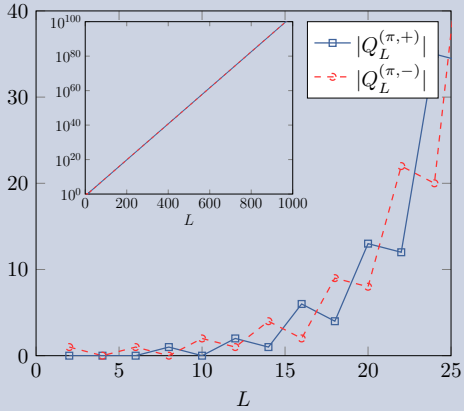
  Number of zero-energy eigenstates in the PXP model
  Buijsman, PRB 2022

  Gaussian state approximation of quantum many-body scars
  Buijsman, Bar Lev, arXiv:2307.11816 (2023)

- $H = \sum_i P_i X_{i+1} P_{i+2}, \qquad P_i = \frac{1}{2}(\mathbb{1} - \sigma_i^z), \qquad X_i = \sigma_i^x$
- Constrained Hilbert space: no two consecutive up-spins
- Particle-hole symmetry: symmetry-protected nullspace

$$H = \begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix}$$

- Dimensionality of nullspace: difference between numbers of "even" and "odd" basis states



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  Buijsman, PRB 2022

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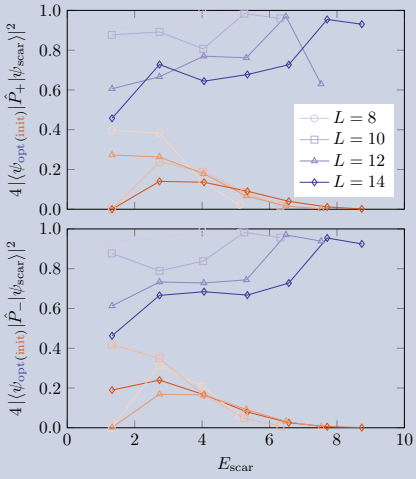
- Quantum many-body scars: low-entangled eigenstates
- Gaussian state approximation: ground state of

$$H = \sum_{i,j} \left[A_{ij} c_i^\dagger c_j + \tfrac{1}{2}\left(B_{ij} c_i^\dagger c_j + B_{ij}^\dagger c_i c_j\right) \right]$$

- Numerical optimization of parameters A_{ij} and B_{ij}

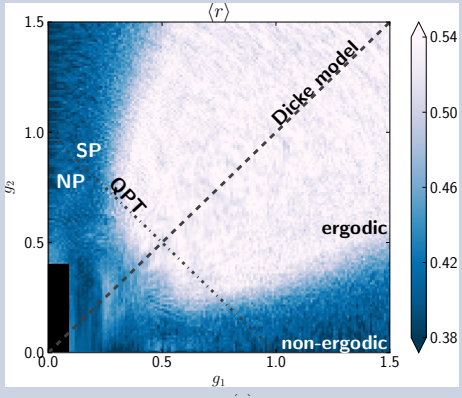


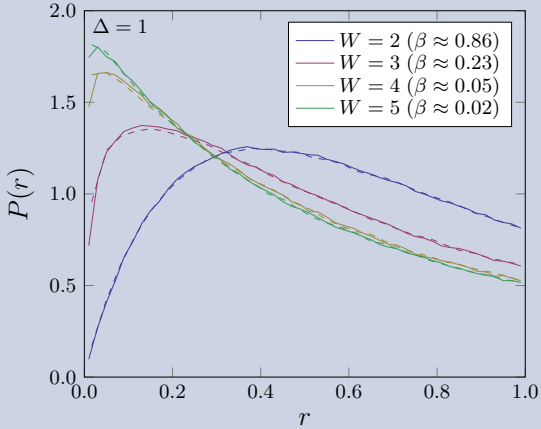
Appendix: some results on other topics

- **Other topics (longer ago)**

  Random matrix ensemble for the level statistics of many-body localization
  Buijsman, Cheianov, Gritsev, PRL 2018

  Non-ergodicity in the anisotropic Dicke model
  Buijsman, Gritsev, Sprik, PRL 2017





– thank you for your attention –